ContextWorld: Benchmarking In-Context Dynamics Adaptation in Latent Visual World Models

Abstract

Latent world models predict future states from visual observations and action sequences within a learned representation space. When environment dynamics vary, accurate prediction requires inferring hidden dynamic properties from interaction history. However, because models can exploit immediate visual shortcuts or regress to average dynamics to achieve low loss without history conditioning, conventional prediction accuracy fails to reveal genuine adaptation. We introduce ContextWorld, a benchmark designed to evaluate in-context dynamics adaptation in action-conditioned latent world models under strictly frozen weights. ContextWorld employs paired counterfactual evaluations—varying interaction histories while holding the current observation and future action sequence fixed—to concurrently quantify adaptation accuracy (whether predictions correctly distinguish between diverging future states) and history sensitivity (whether predictions respond to altered context and yield lower error than mismatched history). Spanning nine tasks across four environments, the benchmark provides separate training and evaluation splits with continuous interaction rollouts under differing dynamics. Our systematic evaluation of representative latent world models (LeWM, PLDM, and DINO-WM) yields three key findings: First, models trained under fixed dynamics make negligible use of interaction history across most tasks. Second, mixing multi-dynamics data improves history responsiveness, but the benefits of data scaling and the relative effectiveness of training strategies vary substantially across tasks. Third, under certain training schemes, models fail to effectively leverage history even on the training distribution, despite steadily decreasing training loss. These results reveal a notable mismatch between prediction-loss optimization and in-context dynamics adaptation, suggesting that standard prediction objectives do not reliably drive models to use interaction history and highlighting the need for objectives and benchmarks explicitly targeted at in-context dynamics adaptation.

1. Introduction

Visual world models learn environment dynamics to predict future states [19], with latent world models performing these predictions within learned representations of visual observations [20, 21]. A model trained under fixed dynamics can achieve high prediction accuracy without utilizing interaction history to adapt its predictions to current conditions. However, such predictions can fail when environment dynamics vary [33]. For example, visually identical objects can move different distances under the same push if surface friction differs. Interaction history provides evidence about these hidden physical properties. We investigate whether latent world models can leverage this contextual evidence to predict future states under varied dynamics *without updating their weights*—an ability we term *in-context dynamics adaptation*.

Physical prediction benchmarks evaluate whether observed interactions help models infer hidden properties or predict downstream outcomes governed by them [4, 32]. While benchmarks like Physion++ assess physical understanding via auxiliary readout classifiers, we introduce ContextWorld to evaluate in-context dynamics adaptation directly within action-conditioned latent world models, leveraging their existing latent predictors without training separate classifiers for future events. The benchmark spans nine tasks across four environments, covering physical parameter variations and interaction rules, and provides separate training and evaluation data.

In its paired tasks, two dynamics conditions share the identical current observation and future action sequence, but produce diverging ground-truth futures. Interaction history is therefore strictly necessary to distinguish the two outcomes. Because a deterministic model that ignores history receives identical inputs and yields identical predictions for both conditions, it scores at most 50% on paired future selection when ties count as failures. This paired counterfactual design restricts history-agnostic heuristics, enabling us to concurrently evaluate two aspects directly in latent space: (1) adaptation accuracy, which measures whether a model's prediction correctly aligns with its corresponding true future rather than the alternative condition's future; and (2) history sensitivity, which measures whether predictions re-

liably respond to altered context, verified by substituting the history with that from an alternate dynamics setting to examine if matching history yields strictly lower error.

Using ContextWorld, we benchmark three representative latent world models: LeWM [24], PLDM [30], and DINO-WM [37]. We find that after training under fixed dynamics, these models make negligible use of interaction history across most tasks. To examine how this adaptation ability can be acquired, we mix ContextWorld training data with original rollouts and compare training from scratch against fine-tuning models pretrained under fixed dynamics. We further scale up the training data and examine the impact of freezing the visual encoder. Our experiments reveal that while data scaling improves history utilization on several tasks, the optimal model and training strategy vary substantially by task. For example, starting from the same pretrained weights, freezing PLDM's visual encoder improves its ability to distinguish future states under varying contact friction, but reduces this ability under motion damping.

To understand why models often fail to leverage history, we analyze model predictions on the training distribution itself. Under certain training schemes, training loss steadily decreases or approaches convergence even while models fail to use interaction history to distinguish physical conditions. Because this failure occurs on the training data, it cannot be explained by out-of-distribution generalization alone, suggesting that standard prediction objectives may not inherently provide sufficient incentive for models to condition their predictions on interaction history.

Our main contributions are:

- **The ContextWorld benchmark and paired evaluation protocol.** We introduce a standardized framework featuring separate training and evaluation data across nine hidden-dynamics tasks. This paired design quantifies both adaptation accuracy and history sensitivity directly from action-conditioned latent rollouts.
- **Empirical findings on adaptation limits.** We benchmark representative models across data scales, training strategies, and visual encoder freezing schemes, demonstrating that optimal strategies are highly task-dependent. Furthermore, we show that training loss can steadily decrease even when models fail to leverage history on the training data, revealing a disconnect between standard prediction losses and in-context dynamics adaptation.

2. Related Work

Visual world models. Visual world models learn environment dynamics to simulate and predict future states. Early frameworks [19] combined learned visual representations with recurrent dynamics to support internal policy training. Subsequent methods advanced this paradigm through online planning (PlaNet [20]) and policy learning via imagined latent trajectories (Dreamer [21]). While these early approaches rely on pixel-level image reconstruction to learn representations, Joint-Embedding Predictive Architectures (JEPAs) [23] shift the training objective toward predicting target representations directly without reconstructing pixels. This principle has been applied to masked spatial regions (I-JEPA [2]) and extended to temporal video representation learning (V-JEPA [5]). To support robotic planning, V-JEPA 2 [3] introduces an action-conditioned predictor trained on robot interactions atop pretrained video representations.

Action-conditioned latent world models also differ in how they obtain their visual representations. DINO-WM [37] trains a predictive dynamics model on frozen pretrained features for planning from offline trajectories. In contrast, PLDM [30] jointly optimizes the visual encoder and the predictor using reward-free offline data. To stabilize joint training, LeWM [24] simplifies the objective to a latent prediction loss and a single representation regularizer. We benchmark these three paradigms to evaluate how structural choices influence in-context dynamics adaptation.

In-context adaptation. When environment dynamics differ between training and deployment, interaction history provides a signal for adaptation without requiring weight updates. Meta-reinforcement learning [6] leveraged this principle by training agents to adapt from limited experience across tasks. RL^2 [14] implicitly stores interaction history within a recurrent network's hidden state, while PEARL [29] infers a task representation from experience to condition the policy. Paralleling large language models, which learn from in-context examples without parameter updates [8, 13], this adaptation paradigm has been extended to visual world models.

Within world models, context guides predictions of action outcomes in new environments. OSWM [16] pretrains a transformer on synthetic dynamics, using observed state transitions as context to forecast futures. L2World [33] distinguishes using context to identify previously seen environments from learning new dynamics. It studies the effects of context length and training environment diversity through multi-step prediction errors and context perturbations. For robotic control, ICWM [34] employs task-agnostic interaction history to adapt policies to novel camera viewpoints without weight updates, using ablations over correct, absent, and mismatched contexts to validate history reliance.

Benchmarks. To evaluate adaptation across environments, existing benchmarks introduce controlled variations in dynamics, appearance, or task rules. Meta-World [35] and Procgen [11] test generalization across manipulation tasks and procedurally generated levels. CausalWorld [1] and CARL [7] vary environment properties, while the XLand series (XLand-MiniGrid [26], XLand-100B [27])

evaluates policy adaptation from interaction history across distributions of tasks.

For physical prediction, several benchmarks focus on counterfactual reasoning and hidden property inference. CoPhy [4] asks models to predict outcomes after changes to a scene's initial configuration, using the original trajectory as evidence of hidden properties. Filtered-CoPhy [22] ensures the observed trajectory contains sufficient information for the counterfactual prediction. ComPhy [9] probes reasoning regarding hidden mass and charge through predictive and counterfactual video querying. Physion++ [32] tests whether interaction history helps predict future contact in paired trials that share the same current observation but differ in physical properties, assessing this understanding via auxiliary readout classifiers.

Beyond visual domains, a recent time-series benchmark [36] reveals that low prediction error can coexist with incorrect response directions when actions change. This echoes our finding of a mismatch between prediction-loss minimization and true dynamics adaptation. A remaining question is when action-conditioned latent world models learn to leverage history for accurate prediction. ContextWorld addresses this gap through controlled comparisons across models, tasks, data scales, and training strategies.

3. ContextWorld Benchmark

ContextWorld is designed to evaluate in-context dynamics adaptation—specifically, whether models use interaction history to accurately predict future states conditioned on an action sequence. This section formalizes the prediction setting, describes how we construct the tasks and datasets, and explains the paired evaluation protocol and metrics.

3.1. Problem Formulation

To isolate in-context adaptation from weight updates, model parameters θ remain fixed during evaluation. The model receives exclusively visual observations and actions. For a target horizon τ, the model receives an interaction history H_t, comprising interleaved visual observations up to time t and their executed actions, alongside a future action sequence $A_{t:t+\tau-1}$. The prediction and its target are

$$\begin{aligned} \hat{z}_{t+\tau} &= f_\theta(H_t, A_{t:t+\tau-1}), \\ z_{t+\tau} &= e_\theta(o_{t+\tau}). \end{aligned} \tag{1}$$

Here, $o_{t+\tau}$ is the ground-truth observation at time $t+\tau$, and e_θ is the model's target encoder, which is also frozen. Predictions are evaluated directly against these encoded targets in latent space.

3.2. Task Design

To evaluate dynamics adaptation, we construct nine tasks spanning four base environments by introducing controlled variations in physical parameters and interaction rules. These environments build upon TwoRoom navigation [37], the PushT pushing environment [10], Reacher from the DeepMind Control Suite [31], and Cube from OGBench [28]. The tasks and their history requirements are summarized in Table 1. Importantly, models are trained separately for each task, and baselines like DINO-WM keep their pretrained visual encoder frozen throughout.

To isolate the role of interaction history, we construct paired counterfactual samples for six tasks: pusher displacement, arm mass, contact friction, motion damping, gripper carry rule, and portal exit. Within each pair, the two samples share the same current observation and future action sequence, but operate under different hidden dynamics. The history and ground-truth future for each sample are generated through one continuous simulation with fixed dynamics. Because the resulting futures differ physically, accurate prediction necessitates conditioning on the interaction history. Trajectory pairing and validity checks are detailed in Appendix B.

Unlike the paired counterfactuals, the remaining three tasks evaluate history usage across multiple condition values or specific environmental rules. The speed task tests generalization to unseen velocities: we hold the current observation and planned actions fixed, providing histories generated at varying speeds to observe if the model scales its predicted motion. Action delay introduces variable latency to test predictions before and after an action takes effect. For example, delays of six or ten simulator steps both result in zero movement within the next five steps. Because these yield identical future states, they are scored as a single response group, with all groups weighted equally (Appendix C). Finally, the door traversability task tests whether the model correctly infers passage or blockage from preceding interactions.

3.3. Dataset Construction and Splits

To evaluate whether models infer dynamics from context rather than simply recalling training trajectories, all tasks provide trajectory-disjoint Training, Development, and Test splits. The dataset design covers two facets of dynamics adaptation: (1) **in-context mode disambiguation**, where paired physical-parameter tasks such as friction, mass, and damping share parameter settings across splits but evaluate models on unseen counterfactual trajectories; and (2) **continuous parameter generalization**, exemplified by Speed, which evaluates interpolation and extrapolation to speeds held out during training. Each paired task provides up to 32,768 training pairs, with 256 balanced pairs in each of Development and Test. Development supports method and checkpoint selection; Test is reserved for final evaluation. Comprehensive dataset statistics, dynamics settings, and split checks appear in Table 4 and Appendix D.

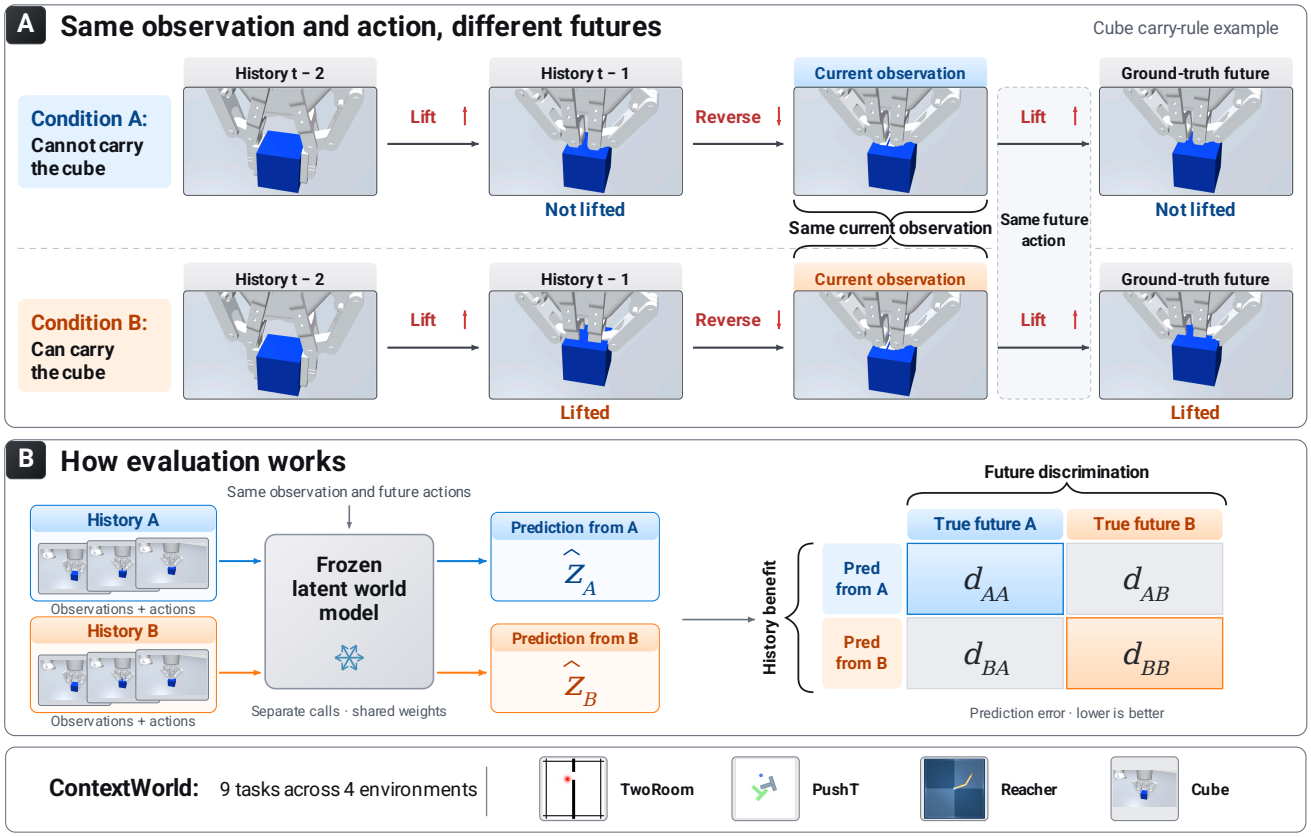


Figure 1. **Controlled prediction and history comparisons.** (A) Cube carry-rule example, re-rendered for clarity. A lift command and its reverse reveal whether the cube is carried, then return both conditions to the same current observation. Repeating the lift produces different futures. Commands label transitions. (B) The frozen model receives observations and actions and produces latent predictions. Rows of the error matrix compare futures for a fixed prediction; columns compare histories for a fixed target.

Table 1. The nine tasks test whether interaction history helps models predict the effects of hidden dynamics. History length h counts observation frames, including the current frame; sampling details are given in Appendix A.

Task	Environment	h	What history helps the model predict
Speed	TwoRoom	3	How far the agent moves under the same action
Pusher displacement	PushT	3	Whether the same action moves the pusher a shorter or longer distance
Arm mass	Reacher	3	How the same torque changes arm motion
Action delay	TwoRoom	7	When an action takes effect
Contact friction	PushT	3	How friction changes object motion during contact
Motion damping	PushT	3	How quickly object motion decays after contact ends
Gripper carry rule	Cube	3	Whether a closed gripper carries the cube
Door traversability	TwoRoom	3	Whether a visually identical door permits passage
Portal exit	TwoRoom	3	Where the agent emerges after entering the same portal

3.4. Evaluation Protocol and Metrics

Evaluating latent in-context dynamics adaptation requires disentangling genuine context utilization from predictive shortcuts. In learned latent spaces, models can achieve low prediction loss by averaging over dynamics conditions or minimizing prediction errors shared across dynamics conditions, without adapting to the underlying physical properties. To rigorously evaluate models under strictly frozen weights, we establish an evaluation protocol structured around four complementary questions: (1) Does the prediction distinguish diverging physical futures? (2) How

accurate is the multi-step trajectory rollout? (3) Does the supplied history reliably reduce prediction error? and (4) Is the differential response to varying dynamics accurately recovered?

Adaptation Accuracy (A_{future}). To test whether predictions distinguish diverging physical outcomes, we evaluate paired counterfactual predictions against latent ground-truth targets. For pair i in a set of N pairs, with latent targets z_i^A, z_i^B and predictions $\hat{z}_i^A, \hat{z}_i^B$, a prediction is scored as correct only if it is strictly closer in MSE (d) to its corresponding future than to the alternative: $I_i^A = \mathbf{1}[d(\hat{z}_i^A, z_i^A) < d(\hat{z}_i^A, z_i^B)]$, with I_i^B defined symmetrically. The paired adaptation accuracy is:

$$A_{\text{future}} = \frac{100}{2N} \sum_{i=1}^{N} \left(I_i^A + I_i^B \right). \tag{2}$$

Ties count as failures; thus, any deterministic model that ignores history receives identical inputs and scores at most 50%. For multi-condition tasks, Speed requires the matching history to yield lower error than all alternative speed contexts, while Action Delay evaluates classification across six latency response groups (Appendix E).

Multi-Step Prediction Error (E/B) and History Controls. Because high classification accuracy does not guarantee low trajectory error—as even minute perturbations toward the correct target can yield 100% accuracy (Appendix F)—we measure complete autoregressive rollouts across up to 25 simulator steps without intermediate ground-truth observations. We normalize the aggregate rollout error across scenes by a reference that does not distinguish dynamics conditions [25]:

$$E/B = \frac{\sum_q L_q}{\sum_q B_q}. \tag{3}$$

For scene q, L_q denotes the mean squared prediction error averaged over K_q dynamics conditions, C_q shared action sequences, and $T = 5$ rollout steps, while B_q is the corresponding error of the mean-future reference $\bar{z}_{qct} = K_q^{-1} \sum_k z_{qkct}$. For equally weighted conditions sharing the current observation and future actions, this mean future minimizes squared error among predictions that do not distinguish the conditions [18]; consequently, any deterministic model that ignores history cannot achieve $E/B < 1$. Lower E/B is better, with $E/B = 0$ denoting exact prediction.

To isolate the effect of the supplied history on prediction error, we swap histories across dynamics conditions while keeping the current observation, future actions, and target fixed. We report **History (%)** (the win-rate where matched history yields strictly lower error than mismatched history) and the normalized error reduction $G_H = (E_- - E_+)/B$, where E_- and E_+ are aggregate errors under mismatched and matched histories, respectively, and $B = \sum_q B_q$.

Normalized Response Error (NRE). Prediction loss can decrease as errors shared across conditions shrink, even when the predicted difference between conditions remains inaccurate. To isolate the model's conditional response to physical differences, we define the Normalized Response Error (NRE) over true difference vectors $\Delta z_i = z_i^B - z_i^A$ and predicted differences $\Delta \hat{z}_i = \hat{z}_i^B - \hat{z}_i^A$:

$$\text{NRE} = \frac{\sum_i \|\Delta \hat{z}_i - \Delta z_i\|^2}{\sum_i \|\Delta z_i\|^2}. \tag{4}$$

NRE $= 0$ denotes an exact differential response, whereas NRE $= 1$ is the baseline of predicting zero difference. Subtracting predictions removes condition-invariant errors, measuring how accurately the model predicts the difference between dynamics conditions.

Evaluation Boundaries. Both E/B and NRE are evaluated in each checkpoint's target representation, not a shared physical space. Changes in that representation must also be considered when comparing training schemes or checkpoints. We report means and standard deviations across runs, with multi-step trajectory intervals estimated via source-cluster bootstrap resampling (Appendix E). Retention of control capabilities in the original environments is evaluated separately via CEM planning (Section 4).

4. Experiments

4.1. Experimental Setup

We evaluate three latent world models with different representation-learning strategies: LeWM [24] combines latent prediction with representation regularization; PLDM [30] jointly learns visual representations and latent dynamics; and DINO-WM [37] trains its predictor on frozen DINOv2 features. All weights remain frozen during evaluation.

Table 2 defines four schemes for testing how data composition, pretraining, and visual encoder updates affect adaptation. Original uses the fixed-dynamics datasets released with LeWM; Mixed samples equal numbers of trajectory windows from those datasets and ContextWorld to support adaptation while retaining original-environment performance. Comparing Mixed with Original $\rightarrow$ Mixed tests the benefit of pretraining. Within the latter, Joint and Frozen Encoder start from the same checkpoint to test whether updating the visual encoder helps. Mixed-data training is performed separately for each task. CEM planning [12] in the original environments measures capability

retention. Optimization and evaluation settings appear in Appendix G.

Table 2. Training schemes. Mixed samples original-environment and ContextWorld trajectory windows equally.

Scheme	Start	Data	Encoder
Original	Initial	Original	Updated†
Mixed	Initial	Mixed	Updated†
Original → Mixed (Joint)	Original	Mixed	Updated
Original → Mixed (Frozen Encoder)	Original	Mixed	Frozen

Initial weights are random for LeWM and PLDM. DINO-WM uses a pretrained visual encoder and a random predictor. †Its encoder is always frozen, so Joint does not apply. Other modules remain trainable; native training losses are retained.

4.2. Adaptation Results

Models trained on fixed-dynamics data show limited ability to distinguish futures under different dynamics: Original scores near or below 50% on the seven binary tasks (Table 3). Adding ContextWorld data improves several tasks, including arm mass for LeWM and PLDM, but does not produce uniformly strong adaptation.

Model advantages also depend on the task. With Mixed training, LeWM and PLDM achieve near-perfect action-delay accuracy, whereas DINO-WM scores 16.65%. On pusher displacement, DINO-WM leads at 97.66%, while LeWM and PLDM remain much closer to the 50% reference. All three reach near-perfect door-traversability accuracy. These rank reversals motivate evaluation across diverse dynamics rather than a single representative task.

Strong one-step adaptation does not establish accurate multi-step prediction. On pusher displacement, LeWM and PLDM trained with Original → Mixed (Frozen Encoder) exceed 98% History win-rate and substantially reduce NRE (Appendix H). Yet Figure 2 shows that their prediction errors on extended action sequences exceed the corresponding mean-future reference at later horizons, despite improving on Original. Aggregate E/B remains above 1.4 for both models. These queries extend both the actions and prediction horizon relative to the one-step tasks; the contrast does not isolate a single cause of rollout error.

4.3. Effects of Training Choices

Freezing the pretrained visual encoder helps some tasks and hurts others. For PLDM, it improves contact friction Test accuracy from 50.00% to 73.63% relative to joint training, but reduces motion damping accuracy from 79.88% to 54.49% (Table 3). Each comparison holds the data, initialization, and training duration fixed. Because projectors and action encoders remain trainable, this intervention does not isolate a static target representation.

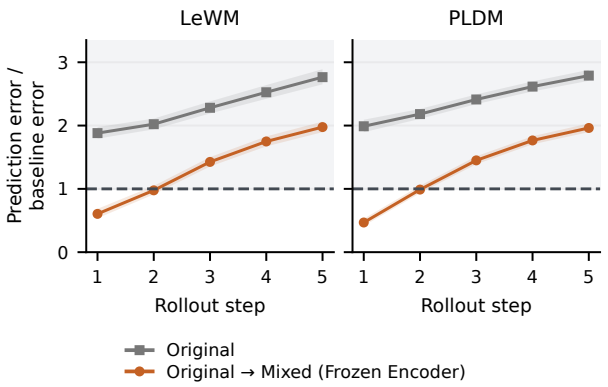


Figure 2. **Multi-step prediction on pusher displacement (Development).** Each rollout step spans five simulator steps without new observations. The dashed line is the reference error ratio (1), obtained by predicting the mean future representation across dynamics conditions. Ratios are computed within each checkpoint's representation, then averaged over runs. Bands give pointwise 95% source-bootstrap intervals. These extended queries differ from the one-step tasks; nine-task summaries appear in Appendix H.

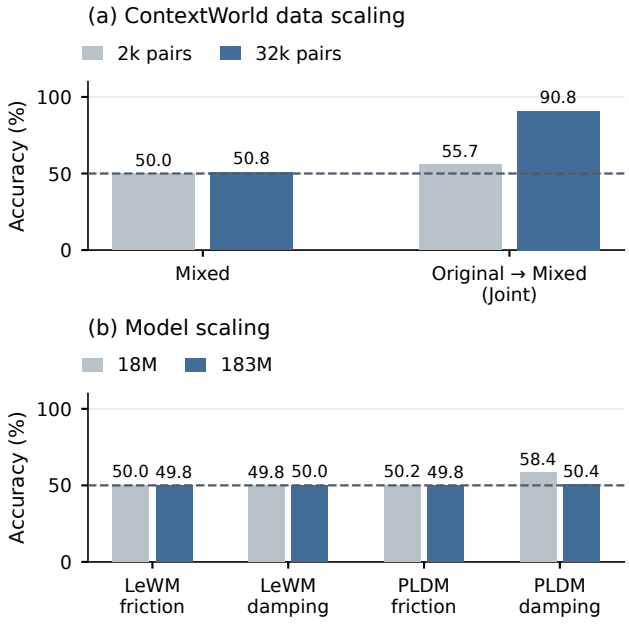


Figure 3. **Data and model scaling on Development.** (a) LeWM on pusher displacement, with 2,048 versus 32,768 ContextWorld training pairs and the same original-environment dataset. (b) LeWM and PLDM under Mixed. Labels above the bars give accuracy (%). Dashed lines mark the 50% paired-task reference. Epoch counts are fixed. Full results and run counts appear in Appendix H.1–H.2.

Figure 3 examines whether more data or capacity resolves weak adaptation. For LeWM on pusher displace-

Table 3. Test task accuracy (%, higher is better). Entries report mean accuracy. Training-run counts and available standard deviations are given in Table 6. Speed reports unseen speeds within the training range; action delay uses seven-frame history. A dash denotes no comparable checkpoint. Each task uses its own checkpoint. Bold marks the highest displayed mean per task, not statistical significance.

Model	Training scheme	Speed	Pusher displacement	Arm mass	Action delay	Contact friction	Motion damping	Gripper carry rule	Door traversability	Portal exit
LeWM	Original	58.04	48.96	50.07	—	49.93	50.00	52.21	50.33	50.07
LeWM	Mixed	98.19	50.78	85.74	97.70	50.00	50.00	50.00	99.83	50.78
LeWM	Original → Mixed (Joint)	96.89	91.80	87.30	—	50.39	50.00	50.00	99.83	50.39
LeWM	Original → Mixed (Frozen Encoder)	**98.44**	97.46	77.15	—	**86.13**	53.91	**76.37**	**100.00**	77.93
PLDM	Original	48.41	45.57	51.82	—	50.00	50.26	50.78	50.06	50.07
PLDM	Mixed	96.78	56.25	**91.02**	**100.00**	50.00	58.20	49.80	**100.00**	72.27
PLDM	Original → Mixed (Joint)	98.33	59.18	84.77	—	50.00	**79.88**	50.20	**100.00**	52.54
PLDM	Original → Mixed (Frozen Encoder)	96.44	**98.63**	81.84	—	73.63	54.49	52.93	**100.00**	91.21
DINO-WM	Original	57.11	38.48	50.78	—	50.39	49.80	50.00	50.00	50.78
DINO-WM	Mixed	59.41	97.66	78.32	16.65	84.18	56.45	60.16	**100.00**	**100.00**
DINO-WM	Original → Mixed (Frozen Encoder)	84.78	—	—	—	—	55.27	62.89	—	—

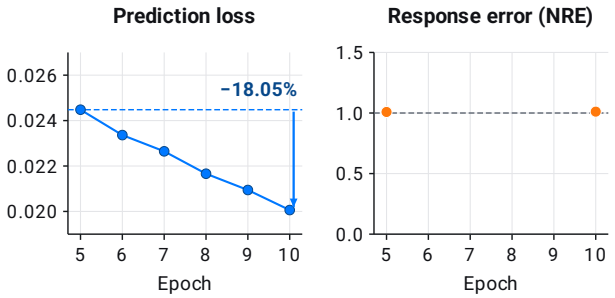


Figure 4. **Lower prediction loss does not ensure an accurate conditional response.** PLDM on contact friction, Original → Mixed (Joint). Recorded prediction loss falls by 18.05% from epochs 5 to 10, while training-query NRE stays near the zero-response reference of 1. Loss uses mixed training batches; NRE is measured on fixed paired training queries at the two endpoints. Full scheme and task comparisons appear in Figure 5.

ment, data scaling substantially benefits Original → Mixed (Joint), whereas Mixed stays near chance. NRE also improves with joint fine-tuning (Appendix H.1). Fixed epochs allow update counts to change with data size, so this is not a fixed-compute comparison.

Under the 10-epoch recipe, expanding LeWM and PLDM from 18M to 183M parameters yields no consistent accuracy gain. PLDM on damping loses 8.01 percentage points of Development accuracy and 10.67 points of original-environment CEM success; CEM success decreases in all four comparisons (Appendix H.2). The cause of this regression remains unresolved.

4.4. Prediction Loss and Adaptation

Weak evaluation performance may reflect poor generalization or conditional responses that were not learned well during training. We compare prediction loss with NRE on fixed training queries at epochs 5 and 10 (Figure 4).

On friction, PLDM joint training reduces prediction loss by 18.05%, yet training-query NRE remains near the zero-response reference. On damping with the encoder frozen, loss approaches convergence while NRE improves but remains high (Figure 5). Changing the scheme helps: freezing yields a stronger conditional response on friction, whereas joint training performs better on damping, with Development showing the same task-dependent ordering.

These results expose a disconnect between prediction-loss optimization and history-conditioned adaptation, even on training data. Prediction loss aggregates errors across mixed data, whereas NRE directly measures error in the predicted differences between conditions. Optimizing the former does not ensure an accurate conditional response. This finding identifies insufficient constraints on history-conditioned dynamics in the evaluated training objectives and strategies.

5. Conclusion

ContextWorld establishes a standardized benchmark to evaluate in-context dynamics adaptation in action-conditioned latent visual world models under strictly frozen weights. Across nine tasks spanning four simulated environments, our paired history comparisons decouple whether models condition their predictions on interaction history from how accurately they forecast the resulting futures. Systematic evaluations of LeWM, PLDM, and DINO-WM demonstrate that while training on multi-dynamics data

can improve adaptation, relative model advantages and the impact of visual encoder freezing remain task-dependent. Crucially, our training-distribution diagnostics expose a disconnect between prediction-loss optimization and in-context adaptation: conditional responses can remain poorly captured even as training loss steadily decreases or approaches convergence. These findings identify insufficient constraints on history-conditioned dynamics within the evaluated training objectives and strategies, underscoring the need for direct adaptation benchmarks and formulations that explicitly constrain history-conditioned responses.

Our findings are framed by specific diagnostic boundaries. Predictions are evaluated in each model's native representation rather than a shared physical error space; most tasks examine adaptation among dynamics conditions represented in training, while Speed evaluates unseen parameter values; and mixed-data checkpoints are trained independently per task, with planning experiments quantifying capability retention in unperturbed environments. These boundaries outline clear avenues for future research: establishing complementary cross-model evaluation in grounded physical coordinates, scaling toward generalist multi-task world models that generalize across broader dynamics, and examining closed-loop planning to test whether history-conditioned latent predictions translate directly into improved decision-making.

References

[1] Ossama Ahmed, Frederik Träuble, Anirudh Goyal, Alexander Neitz, Yoshua Bengio, Bernhard Schölkopf, Manuel Wüthrich, and Stefan Bauer. CausalWorld: A Robotic Manipulation Benchmark for Causal Structure and Transfer Learning. In *International Conference on Learning Representations*, 2021. 2

[2] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-Supervised Learning From Images With a Joint-Embedding Predictive Architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15619–15629, 2023. 2

[3] Mahmoud Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zholus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. *arXiv preprint arXiv:2506.09985*, 2025. 2

[4] Fabien Baradel, Natalia Neverova, Julien Mille, Greg Mori, and Christian Wolf. CoPhy: Counterfactual Learning of Physical Dynamics. In *International Conference on Learning Representations*, 2020. 1, 3

[5] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting Feature Prediction for Learning Visual Representations from Video. *arXiv preprint arXiv:2404.08471*, 2024. 2

[6] Jacob Beck, Risto Vuorio, Evan Zheran Liu, Zheng Xiong, Luisa Zintgraf, Chelsea Finn, and Shimon Whiteson. A Tutorial on Meta-Reinforcement Learning. *Foundations and Trends in Machine Learning*, 18(2–3):224–384, 2025. 2

[7] Carolin Benjamins, Theresa Eimer, Frederik Schubert, Aditya Mohan, Sebastian Döhler, André Biedenkapp, Bodo Rosenhahn, Frank Hutter, and Marius Lindauer. Contextualize Me – The Case for Context in Reinforcement Learning. *Transactions on Machine Learning Research*, 2023. 2

[8] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D. Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language Models are Few-Shot Learners. In *Advances in Neural Information Processing Systems*, 2020. 2

[9] Zhenfang Chen, Kexin Yi, Yunzhu Li, Mingyu Ding, Antonio Torralba, Joshua B. Tenenbaum, and Chuang Gan. ComPhy: Compositional Physical Reasoning of Objects and Events from Videos. In *International Conference on Learning Representations*, 2022. 3

[10] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin CM Burchfiel, and Shuran Song. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. 3

[11] Karl Cobbe, Chris Hesse, Jacob Hilton, and John Schulman. Leveraging Procedural Generation to Benchmark Reinforcement Learning. In *Proceedings of the 37th International Conference on Machine Learning*, pages 2048–2056. PMLR, 2020. 2

[12] Pieter-Tjerk de Boer, Dirk P. Kroese, Shie Mannor, and Reuven Y. Rubinstein. A Tutorial on the Cross-Entropy Method. *Annals of Operations Research*, 134(1):19–67, 2005. 5

[13] Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Jingyuan Ma, Rui Li, Heming Xia, Jingjing Xu, Zhiyong Wu, Baobao Chang, Xu Sun, Lei Li, and Zhifang Sui. A Survey on In-context Learning. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 1107–1128. Association for Computational Linguistics, 2024. 2

[14] Yan Duan, John Schulman, Xi Chen, Peter L. Bartlett, Ilya Sutskever, and Pieter Abbeel. RL^2: Fast Reinforcement Learning via Slow Reinforcement Learning. *arXiv preprint arXiv:1611.02779*, 2016. 2

[15] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993. 12

[16] Fabio Ferreira, Moreno Schlageter, Raghu Rajan, André Biedenkapp, and Frank Hutter. One-shot World Models Using a Transformer Trained on a Synthetic Prior. *arXiv preprint arXiv:2409.14084*, 2024. 2

[17] C. A. Field and A. H. Welsh. Bootstrapping Clustered Data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(3):369–390, 2007. 12

[18] Tilmann Gneiting. Making and Evaluating Point Forecasts. *Journal of the American Statistical Association*, 106(494): 746–762, 2011. 5

[19] David Ha and Jürgen Schmidhuber. Recurrent World Models Facilitate Policy Evolution. In *Advances in Neural Information Processing Systems*, 2018. 1, 2

[20] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning Latent Dynamics for Planning from Pixels. In *Proceedings of the 36th International Conference on Machine Learning*, pages 2555–2565. PMLR, 2019. 1, 2

[21] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to Control: Learning Behaviors by Latent Imagination. In *International Conference on Learning Representations*, 2020. 1, 2

[22] Steeven Janny, Fabien Baradel, Natalia Neverova, Madiha Nadri, Greg Mori, and Christian Wolf. Filtered-CoPhy: Unsupervised Learning of Counterfactual Physics in Pixel Space. In *International Conference on Learning Representations*, 2022. 3

[23] Yann LeCun. A Path Towards Autonomous Machine Intelligence, 2022. Position paper. 2

[24] Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels. *arXiv preprint arXiv:2603.19312*, 2026. 2, 5

[25] Allan H. Murphy. Skill Scores Based on the Mean Square Error and Their Relationships to the Correlation Coefficient. *Monthly Weather Review*, 116(12):2417–2424, 1988. 5

[26] Alexander Nikulin, Vladislav Kurenkov, Ilya Zisman, Artem Agarkov, Viacheslav Sinii, and Sergey Kolesnikov. XLand-MiniGrid: Scalable Meta-Reinforcement Learning Environments in JAX. In *Advances in Neural Information Processing Systems*, 2024. 2

[27] Alexander Nikulin, Ilya Zisman, Alexey Zemtsov, and Vladislav Kurenkov. XLand-100B: A Large-Scale Multi-Task Dataset for In-Context Reinforcement Learning. In *International Conference on Learning Representations*, 2025. 2

[28] Seohong Park, Kevin Frans, Benjamin Eysenbach, and Sergey Levine. OGBench: Benchmarking Offline Goal-Conditioned RL. In *International Conference on Learning Representations*, 2025. 3

[29] Kate Rakelly, Aurick Zhou, Chelsea Finn, Sergey Levine, and Deirdre Quillen. Efficient Off-Policy Meta-Reinforcement Learning via Probabilistic Context Variables. In *Proceedings of the 36th International Conference on Machine Learning*, pages 5331–5340. PMLR, 2019. 2

[30] Vlad Sobal, Wancong Zhang, Kyunghyun Cho, Randall Balestriero, Tim G. J. Rudner, and Yann LeCun. Learning from Reward-Free Offline Data: A Case for Planning with Latent Dynamics Models. *arXiv preprint arXiv:2502.14819*, 2025. 2, 5

[31] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. DeepMind Control Suite. *arXiv preprint arXiv:1801.00690*, 2018. 3

[32] Hsiao-Yu Tung, Mingyu Ding, Zhenfang Chen, Daniel M. Bear, Chuang Gan, Joshua B. Tenenbaum, Daniel L. K. Yamins, Judith Fan, and Kevin A. Smith. Physion++: Evaluating Physical Scene Understanding that Requires Online Inference of Different Physical Properties. In *Advances in Neural Information Processing Systems*, 2023. 1, 3

[33] Fan Wang, Zhiyuan Chen, Yuxuan Zhong, Sunjian Zheng, Pengtao Shao, Bo Yu, Shaoshan Liu, Jianan Wang, Ning Ding, Yang Cao, and Yu Kang. Context and Diversity Matter: The Emergence of In-Context Learning in World Models. In *International Conference on Learning Representations*, 2026. 1, 2, 11

[34] Siyin Wang, Junhao Shi, Senyu Fei, Zhaoyang Fu, Li Ji, Jingjing Gong, and Xipeng Qiu. In-Context World Modeling for Robotic Control. *arXiv preprint arXiv:2606.26025*, 2026. 2

[35] Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. Meta-World: A Benchmark and Evaluation for Multi-Task and Meta Reinforcement Learning. In *Proceedings of the Conference on Robot Learning*, pages 1094–1100. PMLR, 2020. 2

[36] Haochen Zhang, Jiaheng Guo, Zhen Xu, Zachary Plotkin, Nicholas Konz, Zhen Tan, and Tianlong Chen. On the Divergence of Accuracy and Mechanism Consistency in Time Series World Models. *arXiv preprint arXiv:2610.01842*, 2026. 3

[37] Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning. In *Proceedings of the 42nd International Conference on Machine Learning*, pages 79115–79135. PMLR, 2025. 2, 3, 5

A. Input Sampling

Observations are typically sampled every five simulator steps. The actions executed between consecutive observations are included in the interaction history.

B. Trajectory Construction and Pairing

Continuous trajectories. Each sample follows a continuous simulator trajectory from its history through its future, without intermediate resets or state overrides. Identical current observations may still conceal differences in velocity or contact state, so we perform task-specific checks of physical-state matching at the start of prediction. For action delay, we retain the actual queue of pending actions.

Starting positions. For tasks in which the two dynamics conditions cannot naturally share a starting position, we exchange their starting positions during generation and verify that the two conditions have identical sets of starting images. This check reduces the possibility that the starting image alone reveals the condition.

Predictions without history. Consider a complete, balanced pair with distinguishable encoded targets. A deterministic model that ignores history receives the same current observation and future action sequence for both samples and therefore produces the same prediction. This prediction cannot be strictly closer to each sample's own target than to the other target. Consequently, such a model scores at most 50% on the paired future-selection metric in Eq. 2; ties count as failures.

C. Action Delay Response Groups

The delay data retain all 11 delays from 0 to 10 simulator steps. At the five-step horizon used for the main one-step score, delays of 5–10 produce the same stationary response and are grouped together. This score therefore uses six distinguishable response groups: 0, 1, 2, 3, 4, and 5–10.

D. Dataset Statistics and Splits

D.1. Dataset Sizes

Table 4 lists the task data used in the training and evaluation comparisons. Training sizes exclude the original-environment data used for mixed training. A pair contains two trajectories with different dynamics conditions; a trajectory count refers to recorded episodes, not to the number of training windows sampled from them. An evaluation case fixes the current observation and future action sequence and can include histories and targets from several dynamics conditions.

The action delay counts refer to the dataset covering all delays from 0 to 10 simulator steps. Its Training split contains 5,120 base cases, each simulated under all 11 delays, giving 56,320 trajectories. Development and Test each contain 300 base cases, again evaluated under all 11 delays. Door Training contains 7,680 trajectories for each of the passable and blocked rules.

Several examples can share a source episode: pusher displacement can use multiple action windows from one episode, and forward and mirrored damping examples can share a source. The 10,000 gripper carry training pairs use 10,000 distinct source episodes.

D.2. Paired Dynamics Conditions

Pusher displacement uses action gains of 60 and 140. Contact friction uses coefficients 0.05 and 0.80 between the pusher and the object. Motion damping fixes effective contact friction at 0.25 and uses simulator damping settings 0.2 and 1.0; here 0.2 gives faster decay, while 1.0 applies no additional decay. Arm mass uses densities of 500 and 1,500 for the MuJoCo arm link and finger geometry; the hand link retains its original density. Gripper carry changes whether the closed gripper carries the cube. Portal exit uses two exit rules for the same entrance, with paired trajectories reaching the shared query state before entering it. The numeric settings are simulator parameters, not additional model inputs.

D.3. Speed Values and Door Positions

Table 5 lists the speed values used for Training and evaluation. Development and Test use the same evaluation speeds but different interaction trajectories. Each split contains 300 cases for each of the 14 evaluated speeds, or 4,200 cases in total before comparing histories from different speeds. The main task results use the unseen speeds within the training range.

Door position is specified by an integer coordinate along the wall. Test uses the 42 positions $30 + 4k$, where $k = 0, \ldots, 41$. Development uses the 16 positions 65, 77, 79, 87, 112, 115, 131, 152, 167, 169, 171, 172, 175, 180, 183, 185. Training uses the integers from 36 through 188 after excluding all Development and Test positions and the additional unused positions 40, 127, and 132, leaving 96 positions. The three position sets are disjoint. Both passable and blocked rules are used across the splits; the test distinction is the door position, not a new rule.

D.4. Split Checks and Model Inputs

Split checks use source episode identifiers, trajectory template identifiers, or observation hashes, as appropriate for each task. Speed Development uses independently generated initial configurations and excludes initial states and current observations that duplicate Test cases. For Door, we

Table 4. Dataset sizes and condition coverage.

Task	Training	Development	Test
Speed	16,384 trajectories; 32 speeds	300 cases per speed	300 cases per speed
Pusher displacement	32,768 pairs	256 pairs	256 pairs
Arm mass	32,768 pairs	256 pairs	256 pairs
Action delay	56,320 trajectories; 11 delays	300 cases $\times$ 11 delays	300 cases $\times$ 11 delays
Contact friction	32,768 pairs	256 pairs	256 pairs
Motion damping	32,768 pairs	256 pairs	256 pairs
Gripper carry rule	10,000 pairs	256 pairs	256 pairs
Door traversability	15,360 trajectories; 96 door positions	300 cases; 16 door positions	300 cases; 42 door positions
Portal exit	32,768 pairs	256 pairs	256 pairs

Table 5. Speed values in the benchmark. The four evaluation groups are used in both Development and Test. Interpolation values lie within the Training range but are absent from Training.

Setting	Speed values
Training	2.6, 2.7, 2.8, 2.9, 3.1, 3.3, 3.5, 3.7, 3.8, 3.9, 4.1, 4.2, 4.4, 4.5, 4.7, 5.0, 5.1, 5.3, 5.5, 5.7, 5.9, 6.1, 6.2, 6.3, 6.6, 6.7, 6.8, 7.0, 7.2, 7.3, 7.8, 7.9
Seen speeds	3.1, 5.1, 7.0
Interpolation	3.4, 4.8, 6.9
Lower extrapolation	1.75, 1.95, 2.15, 2.35
Upper extrapolation	8.25, 8.75, 9.5, 10.25

also check the position sets listed above for overlap.

Hidden dynamics values, simulator states, and pair identifiers are retained for data generation, validation, and scoring. They are not passed to the prediction model, which receives only the visual observations and actions defined in Sec. 3.1.

E. Evaluation Details

Paired tasks. The strict comparisons in Eq. 2 do not count ties as correct. The two comparisons within a pair are dependent. Appendix B explains the upper bound for a deterministic model that ignores history.

Speed. For each target speed, accuracy is the proportion of cases in which matching history gives strictly lower prediction error than every alternative history. We then average accuracy equally across target speeds. Ties do not count as correct: if all histories produce the same prediction, accuracy is zero. The main results use the interpolation speeds listed in Appendix D.

Action delay. We select the candidate true future with the lowest mean squared error, resolving ties in favor of the smaller delay. The prediction is correct when the selected delay and the true delay belong to the same response group. Within each case, we first average correctness over delays in each group, then average equally across the six groups defined in Appendix C. The final accuracy is the mean over cases.

Door traversability. For a prediction made with matching history, we compare the mean squared errors to the passage and blockage futures. The prediction is correct only if the error to the true future is strictly smaller than the error to the alternative future. Accuracy is averaged over cases and the two rules; ties do not count as correct.

Multi-step prediction horizons. Each prediction step advances five simulator steps. We measure prediction errors at 5, 10, 15, 20, and 25 simulator steps after the current observation.

Within-scene aggregation. Let z_{qkct} and $\hat{z}_{qkct}$ denote the true and predicted future representations for scene q, dynamics condition k, action sequence c, and prediction time t. With K_q conditions, C_q shared action sequences, and $T = 5$ prediction times, the errors used in Eq. 3 are

$$L_q = \frac{1}{K_q C_q T} \sum_{k,c,t} \|\hat{z}_{qkct} - z_{qkct}\|^2,$$

$$B_q = \frac{1}{K_q C_q T} \sum_{k,c,t} \|z_{qkct} - \bar{z}_{qct}\|^2,$$

$$\bar{z}_{qct} = \frac{1}{K_q} \sum_{k} z_{qkct}.$$

L2World tests history dependence by shuffling context observations [33]; our controls instead swap histories between dynamics conditions. For history controls, E_+ and E_- sum the matching- and mismatched-history scene errors using the same averaging over conditions, action sequences, and

prediction times. When a sample has multiple mismatched histories, their errors are averaged equally. A positive G_H indicates that matching history reduces error; it does not imply a low complete prediction error.

Mean-future reference. The reference in Eq. 3 is used only for scoring and is never provided to the model. The ratio is defined only when $\sum_q B_q > 0$. When conditions share the current observation and future actions and are equally weighted, a deterministic model that ignores history cannot achieve $E/B < 1$: it must make the same prediction across conditions, and the mean-future reference minimizes the error of such a prediction.

Response error settings. We specify the prediction horizon for each NRE analysis. The ratio in Eq. 4 is defined only when $\sum_i \|\Delta z_i\|^2 > 0$. NRE equal to 1 does not by itself show that a model ignores history. Errors shared by both predictions cancel in the difference, so an exact conditional response does not imply exact complete predictions.

Uncertainty estimates. For multi-step errors, we use cluster bootstrap resampling [15, 17] at the level of source trajectories or scenes, retaining related samples together. These intervals describe scene-sampling uncertainty for the evaluated checkpoints, not variability across training runs.

Interpreting multiple metrics. The metrics describe different aspects of the same predictions and are not independent pieces of evidence.

F. Scoring Controls

We validate the paired scorer using constructed predictions on 256 Development pusher-displacement scenes, separately in the target representations of LeWM Original and Original → Mixed (Frozen Encoder). The queries use a nonzero action candidate and the five-step prediction horizon; the predictor is not run for this check. Both representations give the results below. With $m = (z_A + z_B)/2$, even predictions $m + 0.001(z_k - m)$ select the correct future while recovering only a tiny fraction of the true response. This control explains why we report response error alongside task accuracy. It verifies the scorer on these queries, not physical calibration across all representations.

Constructed prediction	Accuracy (%)	History (%)	NRE
True target for each condition	100	100	0
Same target z_A for both conditions	50	0	1
Same midpoint m for both conditions	0	0	1
Swapped true targets	0	0	4
$m + 0.001(z_k - m)$ for condition k	100	100	0.998001

G. Training and Diagnostic Settings

Optimization. The training recipe specifies 10 epochs per stage on eight GPUs, with 128 windows per device and no gradient accumulation, giving a global batch size of 1,024. LeWM and PLDM use AdamW with learning rate 5×10^{-5} and weight decay 0.001. DINO-WM uses learning rate 5×10^{-4} and zero weight decay; its pretrained DINOv2-small visual encoder remains frozen. The models retain their respective prediction and regularization losses. The Original → Mixed schemes load model weights but reset optimizer state. Encoder freezing fixes both parameters and training-state buffers; it does not freeze the projector or action encoder.

Input and sampling. The standard setup uses three observation frames, five simulator steps between frames, and a one-frame prediction target. Action delay mixed training instead uses seven-frame history. The original-environment data are the TwoRoom, PushT, Reacher, and Cube datasets released with LeWM; each task uses only the dataset for its environment. Only the ContextWorld Training split is mixed into training. For component lengths N_{orig} and N_{CW}, the mixture has a nominal epoch length of $2\max(N_{\mathrm{orig}}, N_{\mathrm{CW}})$ windows before partitioning and batching, with equal counts from the two components. The smaller component is sampled repeatedly. Consequently, equal epoch counts do not guarantee equal update counts across data scales, and the two-stage schemes include additional original-environment training.

Prediction diagnostics. The main task comparison uses Test. The multi-step and history diagnostics below use Development, with the same checkpoint identities where the input protocol is comparable. Multi-step evaluation covers 2,436 source scenes across the nine tasks and predicts at 5, 10, 15, 20, and 25 simulator steps using only the initial observed history. The candidate actions and multi-step targets extend the one-step task queries. Figure 2 reports E_t/B_t: prediction and mean-future errors are summed over queries separately at each horizon, normalized within each checkpoint, then averaged equally over training repetitions. The aggregate E/B instead combines errors over all horizons before division; it is not the arithmetic mean of these per-horizon ratios. The pointwise intervals use the same source-cluster bootstrap protocol. Each checkpoint is scored separately, and matching training repetitions are then averaged equally. Intervals use 4,000 source-cluster bootstrap samples; all conditions, actions, and times from a source remain together. They condition on the evaluated checkpoints and do not estimate variation from retraining the model. Original action-delay checkpoints do not have the native seven-frame interface and are excluded from this comparison. DINO-WM has no jointly updated visual-encoder scheme; its comparable Original → Mixed results cover Speed, damping, and gripper carry.

Planning evaluation. CEM results measure success in the original environments, with no hidden-dynamics history intervention. Most checkpoint evaluations use six evaluation seeds with 50 episodes each; some original LeWM and PLDM references use three sets of 100 episodes or an aggregated 300-episode result. Evaluation repetitions are pooled within each checkpoint, then training repetitions are averaged equally. A single original-environment checkpoint supplies the reference for all tasks from that environment; these repeated entries are not independent experiments.

Training-query analysis. The friction and damping comparisons use 32,768 training pairs, identical original-environment initialization within each task, and the same 50/50 mixture, batch size, and duration. We evaluate 512 fixed training pairs and 256 Development pairs per task at epochs 5 and 10, predicting the next observation five simulator steps ahead. Loss summaries use the logged mixed-training batches, approximately 50–70 records per epoch, rather than every training batch. Checkpoints and logs are aligned by update step. The curves show recorded means and batch interquartile ranges; loss medians give the same monotonic direction for friction joint training from epochs 5 to 10. NRE is evaluated in each checkpoint's target representation and is not a common physical error across checkpoints. For damping with the encoder frozen, prediction loss changes by only -0.39% between these endpoints and fluctuates in between. The late-training plateau is consistent with the loss approaching convergence.

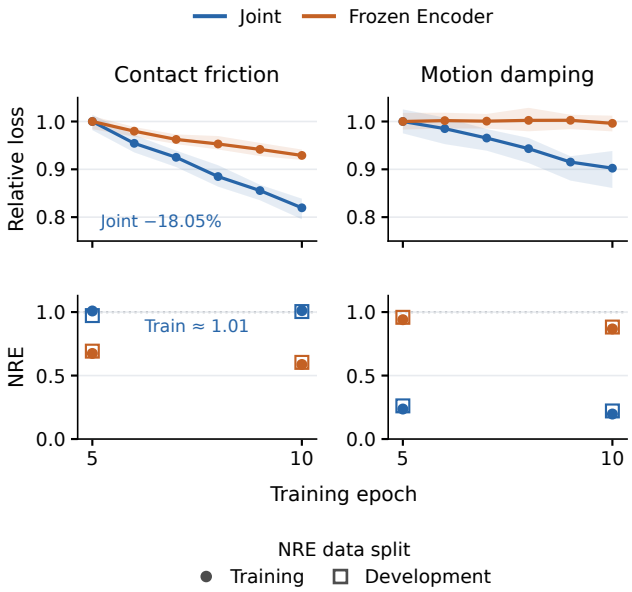


Figure 5. **Training loss and conditional responses across tasks and schemes.** Within each task, PLDM Joint and Frozen Encoder share Original initialization. Loss is normalized to epoch 5; shading shows batch interquartile ranges. NRE contains separate Training and Development measurements at epochs 5 and 10; 1 marks the zero-response reference. Loss uses mixed batches, whereas NRE uses fixed paired task queries.

H. Additional Results

The following tables report variation in Test accuracy across training runs, Development prediction diagnostics, and original-environment planning. History measures strict error reduction from matching history on the task queries; NRE measures the conditional response on those queries. E/B uses the extended multi-step action panels and includes errors shared by both conditions. Because the query sets and prediction horizons differ, these columns should not be treated as measurements of the same event. All latent errors are computed in the checkpoint's own target representation.

Table 6. Training-run variation in Test accuracy. The table lists configurations with multiple training runs; n is the run count and SD is the sample standard deviation in percentage points. All other available entries in Table 3 use one training run, for which sample SD is undefined.

Task	Model	Training scheme	n	Mean (%)	SD (pp)
Speed	LeWM	Original	3	58.04	1.70
Speed	LeWM	Mixed	3	98.19	0.57
Speed	PLDM	Original	3	48.41	4.33
Speed	PLDM	Mixed	3	96.78	2.02
Speed	DINO-WM	Mixed	3	59.41	3.87

Task	Model	Training scheme	n	Mean (%)	SD (pp)
Pusher displacement	LeWM	Original	3	48.96	1.00
Pusher displacement	PLDM	Original	3	45.57	1.88
Arm mass	LeWM	Original	3	50.07	0.11
Arm mass	PLDM	Original	3	51.82	1.30
Action delay	LeWM	Mixed	3	97.70	0.59
Action delay	PLDM	Mixed	3	100.00	0.00
Action delay	DINO-WM	Mixed	3	16.65	0.03
Contact friction	LeWM	Original	3	49.93	0.23
Contact friction	PLDM	Original	3	50.00	0.34
Motion damping	LeWM	Original	3	50.00	0.00
Motion damping	PLDM	Original	3	50.26	0.30
Gripper carry rule	LeWM	Original	3	52.21	1.11
Gripper carry rule	PLDM	Original	3	50.78	0.20
Door traversability	LeWM	Original	3	50.33	0.00
Door traversability	LeWM	Mixed	3	99.83	0.29
Door traversability	PLDM	Original	3	50.06	0.10
Door traversability	PLDM	Mixed	3	100.00	0.00
Door traversability	DINO-WM	Mixed	3	100.00	0.00
Portal exit	LeWM	Original	3	50.07	0.11
Portal exit	PLDM	Original	3	50.07	0.11

Table 7. Development prediction diagnostics and original-environment CEM. Entries average the available training repetitions; E/B intervals are 95% source-bootstrap intervals. History and CEM are percentages. Scheme names match the main text.

Task	Model	Scheme	History ↑	NRE ↓	E/B ↓	E/B interval	CEM ↑
Speed	LeWM	Original	59.15	0.896	1.498	[1.478, 1.517]	92.44
Speed	LeWM	Mixed	98.00	0.042	0.363	[0.345, 0.381]	97.44
Speed	LeWM	Original → Mixed (Joint)	96.67	0.055	0.406	[0.384, 0.429]	97.00
Speed	LeWM	Original → Mixed (Frozen Encoder)	98.56	0.030	0.144	[0.132, 0.156]	95.00
Speed	PLDM	Original	48.48	0.925	3.377	[3.309, 3.445]	90.56
Speed	PLDM	Mixed	97.19	0.084	0.520	[0.498, 0.544]	96.89
Speed	PLDM	Original → Mixed (Joint)	98.89	0.046	0.353	[0.327, 0.379]	96.33
Speed	PLDM	Original → Mixed (Frozen Encoder)	95.56	0.065	0.302	[0.285, 0.319]	98.00
Speed	DINO-WM	Original	58.56	0.928	1.973	[1.954, 1.991]	100.00
Speed	DINO-WM	Mixed	56.56	0.967	2.101	[2.070, 2.133]	99.89
Speed	DINO-WM	Original → Mixed (Frozen Encoder)	86.00	0.411	0.886	[0.847, 0.927]	100.00
Pusher displacement	LeWM	Original	34.05	1.015	2.378	[2.284, 2.481]	82.22
Pusher displacement	LeWM	Mixed	73.63	0.981	3.572	[3.421, 3.741]	78.67
Pusher displacement	LeWM	Original → Mixed (Joint)	97.46	0.222	2.504	[2.425, 2.586]	86.00
Pusher displacement	LeWM	Original → Mixed (Frozen Encoder)	98.05	0.144	1.464	[1.396, 1.537]	73.67
Pusher displacement	PLDM	Original	15.89	1.048	2.471	[2.395, 2.552]	75.67
Pusher displacement	PLDM	Mixed	66.60	0.896	6.533	[6.241, 6.847]	61.33

Task	Model	Scheme	History ↑	NRE ↓	E/B ↓	E/B interval	CEM ↑
Pusher displacement	PLDM	Original → Mixed (Joint)	73.83	0.798	3.811	[3.664, 3.968]	72.00
Pusher displacement	PLDM	Original → Mixed (Frozen Encoder)	99.61	0.115	1.445	[1.388, 1.506]	59.33
Pusher displacement	DINO-WM	Original	14.65	1.113	3.300	[3.166, 3.446]	76.67
Pusher displacement	DINO-WM	Mixed	98.63	0.432	1.996	[1.941, 2.055]	76.67
Arm mass	LeWM	Original	50.98	0.977	687.381	[571.339, 816.332]	55.89
Arm mass	LeWM	Mixed	91.60	0.048	51.291	[45.684, 56.751]	58.00
Arm mass	LeWM	Original → Mixed (Joint)	91.80	0.039	47.152	[41.802, 52.374]	59.00
Arm mass	LeWM	Original → Mixed (Frozen Encoder)	88.28	0.338	31.814	[27.581, 35.547]	59.67
Arm mass	PLDM	Original	66.80	0.922	18.594	[16.630, 20.929]	69.67
Arm mass	PLDM	Mixed	92.19	0.033	13.617	[12.780, 14.631]	51.67
Arm mass	PLDM	Original → Mixed (Joint)	91.02	0.077	44.421	[38.659, 50.208]	58.67
Arm mass	PLDM	Original → Mixed (Frozen Encoder)	92.38	0.211	16.644	[15.067, 18.031]	89.00
Arm mass	DINO-WM	Original	59.38	0.980	48.246	[45.632, 51.198]	90.67
Arm mass	DINO-WM	Mixed	89.26	0.658	14.246	[13.770, 14.655]	75.00
Action delay	LeWM	Mixed	99.00	0.012	1.462	[1.439, 1.487]	87.33
Action delay	PLDM	Mixed	99.99	0.007	1.807	[1.795, 1.820]	90.89
Action delay	DINO-WM	Mixed	21.14	1.000	65.764	[64.835, 66.676]	33.56
Contact friction	LeWM	Original	50.59	1.001	2.753	[2.680, 2.834]	82.22
Contact friction	LeWM	Mixed	50.00	0.999	2.693	[2.642, 2.747]	76.67
Contact friction	LeWM	Original → Mixed (Joint)	57.23	0.999	2.714	[2.674, 2.758]	86.00
Contact friction	LeWM	Original → Mixed (Frozen Encoder)	92.77	0.448	2.891	[2.833, 2.953]	83.33
Contact friction	PLDM	Original	49.28	1.012	3.869	[3.753, 3.992]	75.67
Contact friction	PLDM	Mixed	50.00	1.001	6.269	[5.975, 6.602]	65.67
Contact friction	PLDM	Original → Mixed (Joint)	51.37	1.004	7.086	[6.828, 7.366]	73.33
Contact friction	PLDM	Original → Mixed (Frozen Encoder)	85.94	0.604	4.499	[4.391, 4.618]	79.67
Contact friction	DINO-WM	Original	48.63	1.019	6.109	[5.850, 6.372]	76.67
Contact friction	DINO-WM	Mixed	93.16	0.684	3.673	[3.603, 3.744]	75.67
Motion damping	LeWM	Original	49.09	0.995	5.308	[5.042, 5.616]	82.22
Motion damping	LeWM	Mixed	49.61	1.002	291.614	[253.071, 335.959]	78.00
Motion damping	LeWM	Original → Mixed (Joint)	50.20	1.001	95.787	[83.792, 110.333]	88.33

Task	Model	Scheme	History ↑	NRE ↓	E/B ↓	E/B interval	CEM ↑
Motion damping	LeWM	Original → Mixed (Frozen Encoder)	58.79	0.997	8.202	[7.657, 8.832]	84.33
Motion damping	PLDM	Original	51.69	0.957	5.973	[5.668, 6.318]	75.67
Motion damping	PLDM	Mixed	58.59	1.521	25.424	[22.201, 29.444]	61.00
Motion damping	PLDM	Original → Mixed (Joint)	79.10	0.221	8.247	[7.957, 8.547]	70.00
Motion damping	PLDM	Original → Mixed (Frozen Encoder)	62.11	0.882	5.553	[5.276, 5.871]	78.33
Motion damping	DINO-WM	Original	52.34	1.001	8.465	[8.122, 8.825]	76.67
Motion damping	DINO-WM	Mixed	66.21	1.021	7.273	[7.077, 7.482]	76.67
Motion damping	DINO-WM	Original → Mixed (Frozen Encoder)	65.62	1.083	6.927	[6.731, 7.135]	76.00
Gripper carry rule	LeWM	Original	74.80	0.911	11.678	[10.728, 12.783]	65.44
Gripper carry rule	LeWM	Mixed	49.61	1.001	198.472	[166.622, 241.013]	61.33
Gripper carry rule	LeWM	Original → Mixed (Joint)	52.34	0.998	13.085	[11.613, 14.946]	71.00
Gripper carry rule	LeWM	Original → Mixed (Frozen Encoder)	78.12	0.320	11.937	[11.096, 12.883]	67.00
Gripper carry rule	PLDM	Original	57.62	0.923	15.350	[12.126, 22.152]	53.44
Gripper carry rule	PLDM	Mixed	52.54	0.985	147.591	[121.468, 184.796]	55.00
Gripper carry rule	PLDM	Original → Mixed (Joint)	54.49	0.985	26.553	[24.128, 29.592]	56.67
Gripper carry rule	PLDM	Original → Mixed (Frozen Encoder)	58.01	0.594	29.969	[21.538, 46.388]	56.67
Gripper carry rule	DINO-WM	Original	49.80	1.001	42.857	[41.696, 44.068]	72.67
Gripper carry rule	DINO-WM	Mixed	80.66	0.890	11.789	[11.454, 12.141]	71.67
Gripper carry rule	DINO-WM	Original → Mixed (Frozen Encoder)	82.03	0.874	11.747	[11.417, 12.082]	73.67
Door traversability	LeWM	Original	45.72	0.997	4.676	[4.568, 4.784]	92.44
Door traversability	LeWM	Mixed	100.00	0.045	2.981	[2.938, 3.024]	95.11
Door traversability	LeWM	Original → Mixed (Joint)	100.00	0.047	2.427	[2.378, 2.478]	97.67
Door traversability	LeWM	Original → Mixed (Frozen Encoder)	100.00	0.036	2.442	[2.400, 2.483]	94.00
Door traversability	PLDM	Original	52.22	0.997	7.520	[7.323, 7.718]	90.56
Door traversability	PLDM	Mixed	100.00	0.021	2.404	[2.384, 2.424]	95.78

Task	Model	Scheme	History ↑	NRE ↓	E/B ↓	E/B interval	CEM ↑
Door traversability	PLDM	Original → Mixed (Joint)	100.00	0.027	2.349	[2.302, 2.397]	99.33
Door traversability	PLDM	Original → Mixed (Frozen Encoder)	100.00	0.049	2.455	[2.419, 2.491]	98.67
Door traversability	DINO-WM	Original	61.00	0.986	4.233	[4.143, 4.321]	100.00
Door traversability	DINO-WM	Mixed	100.00	0.223	1.329	[1.303, 1.354]	97.56
Portal exit	LeWM	Original	51.82	0.990	822.254	[587.576, 1303.713]	92.44
Portal exit	LeWM	Mixed	77.54	0.918	58.552	[47.450, 74.465]	98.33
Portal exit	LeWM	Original → Mixed (Joint)	66.21	0.919	61.538	[49.099, 81.373]	95.67
Portal exit	LeWM	Original → Mixed (Frozen Encoder)	93.16	0.253	13.286	[10.625, 17.147]	89.33
Portal exit	PLDM	Original	51.63	0.996	1056.976	[789.611, 1513.078]	90.56
Portal exit	PLDM	Mixed	78.71	0.121	37.556	[35.991, 39.195]	97.00
Portal exit	PLDM	Original → Mixed (Joint)	62.11	0.694	21.881	[20.065, 24.077]	98.67
Portal exit	PLDM	Original → Mixed (Frozen Encoder)	98.24	0.203	10.485	[9.219, 12.101]	98.67
Portal exit	DINO-WM	Original	54.69	0.995	41.991	[40.909, 43.144]	100.00
Portal exit	DINO-WM	Mixed	100.00	0.143	8.092	[8.005, 8.181]	100.00

H.1. Data Scale

These comparisons use Development and hold training epochs fixed. The sample counts describe pairs, not necessarily independent source episodes. Pusher displacement uses 2,048, 8,192, and 32,768 pairs; friction and damping compare 8,192 with 32,768; mass and portal exit compare 2,048 with 32,768. Gripper carry expands from 2,048 to 10,000 source episodes. Portal expansion also broadens initial distances and action magnitudes, so its comparison changes coverage as well as size.

Table 8. Data-scale comparisons on Development. n counts training repetitions contributing to task accuracy and NRE. Planning repetitions can differ and are not counted as additional training runs.

Task	Model	Scheme	Pairs	n	Accuracy (%)	NRE
Pusher displacement	LeWM	Mixed	2,048	3	50.00	1.000
Pusher displacement	LeWM	Mixed	32,768	1	50.78	0.981
Pusher displacement	LeWM	Original → Mixed (Joint)	2,048	1	55.66	0.810
Pusher displacement	LeWM	Original → Mixed (Joint)	8,192	1	82.42	0.390
Pusher displacement	LeWM	Original → Mixed (Joint)	32,768	1	90.82	0.222
Pusher displacement	PLDM	Mixed	2,048	3	50.07	1.003
Pusher displacement	PLDM	Mixed	32,768	1	56.64	0.896
Pusher displacement	DINO-WM	Mixed	2,048	3	89.19	0.583
Pusher displacement	DINO-WM	Mixed	32,768	1	96.09	0.432

Task	Model	Scheme	Pairs	n	Accuracy (%)	NRE
Arm mass	LeWM	Mixed	2,048	3	85.35	0.132
Arm mass	LeWM	Mixed	32,768	1	87.11	0.048
Arm mass	PLDM	Mixed	2,048	3	52.60	1.054
Arm mass	PLDM	Mixed	32,768	1	91.21	0.033
Arm mass	DINO-WM	Mixed	2,048	3	51.37	0.999
Arm mass	DINO-WM	Mixed	32,768	1	80.47	0.658
Contact friction	LeWM	Mixed	8,192	3	50.07	1.000
Contact friction	LeWM	Mixed	32,768	1	50.00	0.999
Contact friction	PLDM	Mixed	8,192	3	50.13	1.001
Contact friction	PLDM	Mixed	32,768	1	50.20	1.001
Contact friction	DINO-WM	Mixed	8,192	3	49.61	1.025
Contact friction	DINO-WM	Mixed	32,768	1	81.84	0.684
Motion damping	LeWM	Mixed	8,192	3	50.07	0.999
Motion damping	LeWM	Mixed	32,768	1	49.80	1.002
Motion damping	PLDM	Mixed	8,192	3	51.89	2.122
Motion damping	PLDM	Mixed	32,768	1	58.40	1.521
Motion damping	DINO-WM	Mixed	8,192	3	50.13	1.014
Motion damping	DINO-WM	Mixed	32,768	1	57.42	1.021
Gripper carry rule	LeWM	Mixed	2,048	3	49.93	1.001
Gripper carry rule	LeWM	Mixed	10,000	1	50.00	1.001
Gripper carry rule	PLDM	Mixed	2,048	3	50.00	0.996
Gripper carry rule	PLDM	Mixed	10,000	1	50.00	0.985
Gripper carry rule	DINO-WM	Mixed	2,048	3	50.33	0.995
Gripper carry rule	DINO-WM	Mixed	10,000	1	61.33	0.890
Portal exit	LeWM	Mixed	2,048	3	50.91	0.925
Portal exit	LeWM	Mixed	32,768	1	51.95	0.918
Portal exit	PLDM	Mixed	2,048	3	69.53	0.155
Portal exit	PLDM	Mixed	32,768	1	73.63	0.121
Portal exit	DINO-WM	Mixed	2,048	3	99.41	0.554
Portal exit	DINO-WM	Mixed	32,768	1	100.00	0.143

H.2. Model Capacity

Both model sizes use the same 32,768-pair datasets, training seed, Mixed initialization, and 10-epoch recipe. The larger model changes the visual backbone from ViT-Tiny to ViT-Base, the representation width from 192 to 768, and predictor depth from six to nine layers. This is a joint expansion of the encoder and predictor, not a predictor-only intervention.

		Accuracy (%)		NRE		CEM (%)	
Task	Model	18M	183M	18M	183M	18M	183M
Contact friction	LeWM	50.00	49.80	0.999	1.000	76.67	74.00
Motion damping	LeWM	49.80	50.00	1.002	1.000	78.00	71.33
Contact friction	PLDM	50.20	49.80	1.001	1.107	65.67	47.33
Motion damping	PLDM	58.40	50.39	1.521	1.536	61.00	50.33

The four endpoint comparisons do not show a consistent improvement from increasing capacity under this recipe. Only measured checkpoints are used in the training-curve analysis: epochs 1–10 for the large LeWM models and epochs 5 and 10 for the other models. There is no shared trajectory of accurate training responses followed by worsening Development response error across these runs. This does not exclude earlier overfitting where checkpoints are unavailable, or gains from retuning the larger models.